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LANDING GEARS FOR AERIAL VEHICLE TO MINIMISE AERODYNAMIC DRAG DURING FLIGHT

Abstract

An aerial vehicle (AV) **100** includes a rotor system **150** with a set of rotor arms **152** and a set of propellers **154** to provide a lift force during a vertical take-off of the AV **100**; and a set of landing gears movably coupled to the rotor arms **152** for movement between a deployed position to support the aerial vehicle **100** during landing and take-off, and a retracted position to form an enclosure **210** around the set of propellers **154** to reduce aerodynamic drag during a forward movement of the AV **100** after the vertical take-off. The sets of landing gears can includes a pair of L-shaped landing gears **200**, a C-shaped cross sectional landing gears, an another C-shaped cross sectional landing gears, a U-shaped slidable landing gears, and a flap-shaped slidable landing gears.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] Benefit is claimed to Indian Patent Application number 202421017458, filed Mar. 11, 2024, the contents of which are incorporated by reference herein in their entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to the field of aerial vehicles. In particular, it pertains to an aerial vehicle. Specifically, the present disclosure pertains to different sets of landing gears of the aerial vehicle.

BACKGROUND

[0003] Background description includes information that may be useful in understanding the present invention. It is not an admission that any of the information provided herein is prior art or relevant to the presently claimed invention, or that any publication specifically or implicitly referenced is prior art.

[0004] Hybrid quadrotor technology offers an innovative and logistically simple solution to problem of vertical take-off and landing (VTOL) by combining the vertical take-off and landing capabilities of a quadrotor, and efficiency, speed, and range of a normal fixed-wing aircraft. Thus, a hybrid aerial vehicle (AV) can take-off and land vertically like helicopters, and transit to cruise flight like airplanes.

[0005] A Typical fixed-wing VTOL AV consists of rotor-blade arms along with fixed wings; and their integration in a single Aerial Vehicle. Besides offering performance advantage of working like a fixed-wing aircraft with vertical take-off and landing performance, they offer other advantages as well like no landing place restrictions, and that of long-range and high-speed characteristics. On account of the above advantages, these aerial vehicles are a versatile choice for a wide range of commercial and defence applications.

[0006] However, these aerial vehicles suffer from high aerodynamic drag when the mode is changed to fixed wing flight and its rotor propellers are not rotating. The presence of stationary propellers can lead to aerodynamic drag. In certain cases, the landing gears are also found to hinder camera view leading to a reduced situational awareness, impaired object detection and tracking. In addition, the landing gears too add to the aerodynamic drag, affecting range and endurance. There is, therefore, a requirement to address the abovementioned drawbacks and limitation of the hybrid VTOL aerial vehicles, so that their full potential could be realized.

[0007] A general object of the present disclosure is to provide an improved aerial vehicle that overcomes drawbacks of the conventional aerial vehicles.

[0008] An object of the present disclosure is to provide an aerial vehicle with reduced aerodynamic drag during flight.

[0009] An object of the present disclosure is to provide improved landing gears sets for aerial vehicles that reduce aerodynamic drag during flight.

[0010] Another object of the present disclosure is to provide improved landing gears for aerial vehicles that do not obstruct field of view (FOV) to the camera during flight.

[0011] Yet another object of the present disclosure is to provide improved landing gears for aerial vehicles that optimizes the aerodynamic profile of the drone to reduce aerodynamic drag during flight.

SUMMARY

[0012] Aspects of the present disclosure relate to the field of aerial vehicles (AV). In particular, the present disclosure relates to landing gears for aerial vehicles, such as hybrid vertical take-off and

landing aerial vehicles, to reduce aerodynamic drag during flight after vertical take-off of the AV. Specifically, the present disclosure provides foldable landing gears that fold after the vertical take-off and take a position to reduce aerodynamic drag on account of rotors that are used for vertical take-off and landing.

[0013] In an aspect, the proposed aerial vehicle includes a rotor system to provide a lift force during a vertical take-off of the aerial vehicle and a set of landing gears. The rotor system includes a set of rotor arms and a set of propellers coupled to the rotor arms. The set of landing gears are movably coupled to the rotor arms for movement between a deployed position, in which the set of landing gears support the aerial vehicle during landing and take-off, and a retracted position, in which the set of landing gears form an enclosure around the set of propellers to reduce aerodynamic drag during a forward movement of the aerial vehicle after the vertical take-off of the aerial vehicle.

[0014] In an embodiment, each of the landing gears may include a pair of L-shaped support members, one on two opposite sides of the corresponding propeller. Each of the support members may have a vertical leg and a horizontal leg, with the horizontal leg of the two support members pivotally coupled to the corresponding rotor arm at a free end of the horizontal leg for movement between the deployed position and the retracted position such that, in the deployed position, a free end of the vertical leg of at least one of the two support members supports the UAV on ground. In the folded position the vertical legs of the two support members may be disposed horizontally under the corresponding propeller to create the enclosure around the propeller.

[0015] In an alternate embodiment of the landing gears, each of the landing gears may have a C-shaped cross section with an upper horizontal portion, a lower horizontal portion and a vertical portion, and the upper horizontal portion, at its free end, may be pivotally coupled to the corresponding rotor arm for movement between the deployed position and the retracted position. In the deployed position the lower horizontal portion supports the UAV on ground, and in the folded position the vertical portion is disposed horizontally under the corresponding propeller to create the enclosure around the propeller.

[0016] In another alternate embodiment of the landing gears, each of the landing gears may have a C-shaped cross section having a pair of vertical portions, and a horizontal portion connecting the two vertical portions. The vertical portions may be in sliding engagement with the corresponding rotor arm for movement between the deployed position, in which the horizontal portion moves down to support the UAV on ground, and the retracted position, in which the horizontal portion moves up to create the enclosure around the corresponding propeller along with the two vertical portions.

[0017] In yet another alternate embodiment of the landing gears, each of the landing gears may include a plurality of U-shaped slats coupled to the corresponding rotor arm at two free ends of the U-shaped slats. The plurality of U-shaped slats may be configured to be moved between the deployed position, in which the plurality of U-shaped slats shrink to move away from the corresponding propeller with a lower end of the U-shaped slats supporting the UAV on ground, and the retracted position, in which the plurality of slats extend to create the enclosure around the corresponding propeller.

[0018] In still another alternate embodiment of the landing gears, each of the landing gears may include a plurality of flaps connected together with at least one joining mechanism system and located under the corresponding rotor. The at least one joining mechanism may be configured to move the plurality of flaps between the deployed position, in which the plurality flaps are oriented vertically with gaps therebetween to allow airflow to pass there through and also support the UAV on ground, and the retracted position, in which the plurality of flaps are oriented horizontally without any gap therebetween to create the enclosure around the corresponding propeller.

[0019] In an embodiment, the aerial vehicle may include a first actuation mechanism to move the set of landing gears between the deployed position and the retracted position.

[0020] In an embodiment, the aerial vehicle may include a second actuation mechanism lock the set of propellers, before the corresponding landing gear is moved to the retracted, such that blades of the propeller are aligned with a flight direction to minimize a frontal area of the enclosure, thereby minimizing aerodynamic drag.

[0021] In an embodiment, the second actuation mechanism may be selected from a group comprising a mechanical actuator mechanism, an electronic actuator mechanism based on sensors, and a magnetic actuator mechanism.

[0022] Various objects, features, aspects and advantages of the inventive subject matter will become more apparent from the following detailed description of preferred embodiments, along with the accompanying drawing figures in which like numerals represent like components.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] The accompanying drawings are included to provide a further understanding of the present disclosure and are incorporated in and constitute a part of this specification. The drawings illustrate exemplary embodiments of the present disclosure and, together with the description, serve to explain the principles of the present disclosure.

[0024] FIG. 1 illustrates a top view of a typical hybrid vertical take-off and landing aerial vehicle, showing major parts of the aerial vehicle.

[0025] FIG. 2A illustrates an exemplary front view of a L-shaped hingedly coupled set of landing gears in a deployed position, in accordance with an embodiment of the present disclosure.

[0026] FIG. 2B illustrates an exemplary front view of the L-shaped hingedly coupled set of landing gears of FIG. 2A in a retracted position, in accordance with an embodiment of the present disclosure.

[0027] FIG. 3A illustrates an exemplary front view of the disclosed C-shaped hingedly coupled landing gears in a deployed position, in accordance with an embodiment of the present disclosure.

[0028] FIG. 3B illustrates an exemplary front view of the C-shaped hingedly coupled landing gears of FIG. 3A in a retracted position, in accordance with an embodiment of the present disclosure.

[0029] FIG. 4A illustrates an exemplary front view of a C-shaped vertically sliding type landing gears in a deployed position, in accordance with an embodiment of the present disclosure.

[0030] FIG. 4B illustrates an exemplary front view of the C-shaped vertically sliding type disclosed landing gears of FIG. 4A in a retracted position covering rotor system, in accordance with an embodiment of the present disclosure.

[0031] FIG. 5A illustrates an exemplary front view of an U-shaped horizontally sliding type fourth disclosed set of landing gears in a deployed position, in accordance with an embodiment of the present disclosure.

[0032] FIG. 5B illustrates an exemplary front view of the U-shaped horizontally sliding type set of landing gears of FIG. 5A in a retracted position covering rotor system, in accordance with an embodiment of the present disclosure.

[0033] FIG. 6A illustrates an exemplary bottom view of a flap-based set of landing gears in a retracted position, in accordance with an embodiment of the present disclosure.

[0034] FIG. 6B illustrates an exemplary front view of the flap-based set of landing gears of FIG. 6A in a deployed position, in accordance with an embodiment of the present disclosure.

[0035] FIG. 6C illustrates an exemplary front view of the flap-based set of landing gears of FIG. 6A depicting a retracted position of the landing gear covering a rotor system, in accordance with an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0036] The following is a detailed description of embodiments of the disclosure depicted in the

accompanying drawings. The embodiments are in such detail as to clearly communicate the disclosure. However, the amount of detail offered is not intended to limit the anticipated variations of embodiments; on the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the present disclosure as defined by the appended claims.

[0037] Embodiments explained herein relate to the field of aerial vehicles (AV). In particular, the present disclosure relates to landing gears for aerial vehicles that reduce aerodynamic drag during flight, such for hybrid vertical take-off and landing aerial vehicles, where drag due to landing gears during flight is more pronounced, thereby making the AVs more promising.

[0038] It is to be appreciated that while the proposed concept of foldable landing gears is especially useful in case of hybrid VTOL aerial vehicles, the same can also be applied to conventional multi-copter aerial vehicles as well, such as to reduce drag due to one or more rotors that may be non-functional for operational reasons, to enhance efficiency of the aerial vehicle.

[0039] In one or more embodiments, the present disclosure provides different configurations of foldable landing gears that, in a deployed position, support the AV on ground, and in the folded/retracted position, when the AV is in flight and rotors used for vertical take-off have been turned off, such as in case of a hybrid VTOL aerial vehicle. make an enclosure around the corresponding rotors to reduce in-flight aerodynamic drag on the AV.

[0040] In an embodiment, by enclosing the rotor within a hollow enclosure during the flight, when the propeller is stationary, a more streamlined shape of the drone is created which guides the airflow around the stationary propellers in a controlled manner, reducing the resistance encountered during forward flight, thus reducing the aerodynamic drag caused by the exposed cross-sectional area of the stationary propellers.

[0041] In an embodiment, formed enclosure by the landing gear in retracted position is hollow such that the hollow enclosure is designed for controlled air flow.

[0042] It is to be appreciated that though, in different embodiments of the proposed landing gear, the landing gears have been identified by alphabetic letters, such as L-shaped, C-shaped and U-shaped, the alphabetic letters do not in any manner limit the shape of the respective landing gears in any manner whatsoever to the corresponding letter, and the landing gears can be of other shapes as well within the concept of their ability to move between a deployed position, in which they support the aerial vehicle, and a retracted position, in which they form an enclosure around the corresponding rotor/propeller.

[0043] FIG. 1 shows an exemplary top view of a typical aerial vehicle depicting various major parts of the AV. The aerial vehicle **100**, which can be an unmanned aerial vehicle (UAV), provides the benefits of multi-rotor platforms to provide transition between the two modes of flight namely a vertical take-off/landing and a level flight.

[0044] The aerial vehicle **100** (also referred simply as AV or UAV, and these are used interchangeably, herein) includes a wing structure **102** configured cross-sectional to a fuselage **104**. A rotor system such as **150-1**, **150-2**, **150-3** and **150-4** (collectively referred to as rotor system **150** herein) includes a set of rotor arms **152**, and a set of top facing propellers **154** coupled to the rotor arm **152**. The propellers **154** are conventional propellers sized according to the AV size, to meet the requirement of vertical take-off and landing. Also, the AV **100** includes one or more front facing propellers **106** for normal flight. The propellers **154** are functional during vertical take-off and landing and stop functioning and may get folded during flight. On the other hand, the front facing propeller **106** is non-functional during vertical take-off and run only during the flight. During the flight, the propellers **154** cause aerodynamic drag as airflow from front directly impinges their rotors and propellers, effecting efficiency of the AVs.

[0045] Furthermore, landing gears (not shown in FIG. 1) hinder camera view due to obstructions caused by the landing gears position, which reduces situational awareness, object detection and tracking. In an aspect, the present disclosure provides a simple, easy to install solution for the

above problems, The present disclosure provides different configurations of landing gears, such as landing gears **200**, **300**, **400**, **500**, and **600**, that conceal propeller system **154** during flight of the aerial vehicle **100** by forming, in a retracted position of the landing gears **200**, **300**, **400**, **500**, and **600**, an enclosure around the propellers **154** to prevent frontal airflow from impinging the corresponding propellers **154**.

[0046] FIGS. **2A** and **2B** show exemplary front views of a first embodiment of the disclosed landing gear (also referred to as L-shaped hingedly coupled landing gear) in deployed position and retracted/folded position respectively.

[0047] In an embodiment, the landing gears **200** includes a pair of L-shaped support members, such as support members **206A** and **206B** (individually and collectively referred to as support member(s) **206**, herein), one on two opposite sides of the corresponding propeller **154**. Each of the support members **206** includes a vertical leg, such as vertical legs **202A** and **202B** (individually and collectively referred to as vertical leg(s) **202**, herein) and a horizontal leg such as horizontal legs **204A** and **204B** (individually and collectively referred to as horizontal leg(s) **204**, herein), with the horizontal leg **204** of the two support members **206** pivotally coupled at hinge joints, such as hinge joint **208-1** and **208-2** (collectively referred to as hinge joint **208**, herein), to the corresponding rotor arm **152** at a free end of the horizontal leg **204** for movement between the deployed position and the retracted position, as shown in FIGS. **2A** and **2B**.

[0048] In an embodiment, as shown in FIG. **2A**, in the deployed position, the free end of the vertical leg **202** of at least one of the two support members **206** supports the UAV on ground, and as shown in FIG. **2B**, in the retracted position, the landing gear **200** forms a hollow cuboid structure around the corresponding propeller providing an enclosure.

[0049] FIGS. **3A** and **3B** show exemplary front views of a second embodiment of the disclosed landing gear **300** (also referred to as C-shaped hingedly coupled landing gear) in deployed position and retracted/folded position respectively.

[0050] In an embodiment, each of the landing gears **300** includes a single member with a C-shaped cross section having an upper horizontal portion **302-1**, a lower horizontal portion **302-2** and a vertical portion **304**. The upper horizontal portion **302-1** is pivotally coupled, at a hinge point **308**, to the corresponding rotor arm **152** at a free end of the upper horizontal portion **302-1** for movement between the deployed position and the retracted position.

[0051] In an embodiment, in the deployed position, the lower horizontal portion **302-2** supports the UAV on ground, as shown in FIG. **3A**, and

[0052] in the retracted position the landing gear **300** covers the corresponding rotor system **150**, as shown in FIG. **3B**. Specifically, in the retracted position, the vertical portion **304** gets reoriented horizontally under the corresponding propeller **154**, as shown in FIG. **3B**. This creates the enclosure **306** around the propeller **154** to reduce aerodynamic drag.

[0053] FIGS. **4A** and **4B** show exemplary front views of third embodiment of the disclosed landing gear **400** (Also referred to C-shaped vertically sliding type landing gears, herein) in deployed position and retracted position respectively.

[0054] In an embodiment, the landing gear **400** includes a C-shaped cross section having a pair of vertical portions **402-1** and **402-2**, and a horizontal portion **404**. The two vertical portions **402-1** and **402-2** are coupled in sliding engagement with a set of rails **406-1** and **406-2** fixed with the corresponding rotor arm **152** to facilitate up and down sliding movement for the landing gear **400**. In the deployed position, the horizontal portion **404** moves down to support the UAV **100** on ground during landing, as shown in FIG. **4A**. In the retracted position of the landing gear **400**, as shown in FIG. **4B**, the landing gear **400** conceals the corresponding propeller **154** by forming an enclosure around the rotor **150**.

[0055] FIGS. **5A** and **5B** show perspective views of a fourth embodiment of the disclosed landing gear **500** (also referred to as U-shaped horizontally sliding type, herein) in the deployed position and retracted position respectively. The landing gears **500** includes a plurality of U-shaped slats,

such as slats **502-1**, **502-1**, . . . , **502-N** (individually and collectively referred to as slat(s) **502**, herein) coupled to the corresponding rotor arm **152** at two free ends, such as free ends **504-1** and **504-2**, of the U-shaped slats **502**.

[0056] In an embodiment, the landing gear **500** can be an origami bellow type retractable system. Specifically, the origami-inspired bellow shaped landing gear **500** can fold along predetermined fold lines, thereby forming a secure enclosure of variable shape and size, such that in the deployed position (as shown in FIG. 5B), when the landing gear is to support the aerial vehicle on ground, the size of the landing gear **500** is smaller, and when the landing gear **500** is retracted (as shown in FIG. 5A) during flight, the size of the landing gear **500** increases for covering the propeller **154**. FIGS. 5A and 5B show a length difference of the landing gear **500** in the deployed and the retracted positions.

[0057] In an embodiment, in the retracted position, the plurality of U-shaped slats **502** are configured to move towards the corresponding propeller **154** to conceal the propeller **154** forming an enclosure, as shown in FIG. 4B, In the deployed position, shown in FIG. 5A, the slats **504** shrink and are moved away from the propeller **154**, allowing the propeller to rotate and provide a lift force. In this position, lower ends **506** of all the U-shaped slats **502** collectively support the UAV **100** on ground, as shown in FIG. 5A.

[0058] In an embodiment, the landing gear **500** can be operatively coupled with a propeller alignment mechanism (also referred to as a second actuation mechanism, herein), such that when the landing gear **500** is retracted to cover the stationary propeller **154**, the landing gear **500** aligns the direction of the propeller **154** along the forward direction of the UAV flight.

[0059] FIGS. 6A and 6B show exemplary bottom views of a fifth embodiment of the disclosed landing gear **600** (also referred to as flap-based horizontally sliding type landing gear, herein) in deployed position and the retracted position respectively.

[0060] In an embodiment, the landing gear **600** includes a plurality of flaps, such as flaps **602-1**, **602-2**, . . . , **602-N** (individually and collectively referred to as flap(s) **602**, herein) configured in a plane below the corresponding propeller and connected together by a joining mechanism, such as the joining mechanism **604-1** and **604-2** (collectively referred to as joining mechanism **604**, herein). The joining mechanism **604** can move the plurality of flaps **602** between the retracted position, in which the flaps **602** are oriented horizontally eliminating gaps **606-1**, **606-2**, . . . , **606-N** (refer FIG. 6A) therebetween to prevent airflow, as shown in FIG. 6B, and form an enclosure. In the deployed position, the flaps **602** are oriented vertically with gap **606** therebetween, which allows airflow from the corresponding propeller **152** to pass through for the propeller **154** to provide the lift force. In the vertical orientation, the flaps **604** can support the AV on ground at a lower edge of the flaps **602**.

[0061] FIG. 6C illustrates an exemplary front view of the UAV with flap-based landing gear **600** in retracted position covering the rotor **152**, in which the plurality of flaps **602** are oriented to create the enclosure **608** around the corresponding propeller **154**.

[0062] In an embodiment, the Aerial vehicle **100** with any of the landing gears **200**, **300**, **400**, **500** and **600** may include a first actuation mechanism, such that the first actuation mechanism actuates the corresponding set of landing gears **200**, **300**, **400**, **500** and **600** to the retracted position when the rotor stops to conceal the set of propellers **154** and form an enclosure.

[0063] In an embodiment, the Aerial vehicle **100** with any of the landing gears **200**, **300**, **400**, **500** and **600** may include a second actuation mechanism which can automatically align the direction of the propellers **154** when the propellers **154** stops along a direction of the flight before the landing gear **200**, **300**, **400**, **500** and **600** is moved to the retracted position such that area of the propeller **154** exposed to the frontal airflow is minimized. This can also help in reducing the size of the enclosure around the propellers **154**, thereby offering least resistance to the frontal airflow and reducing the aerodynamic drag.

[0064] In an embodiment, the second actuation mechanism can be selected at least one from a

mechanical actuator mechanism, an electronic actuator mechanism based on sensors, and a magnetic actuator mechanism where each blade of the propellers **154** can be fitted with magnets of same polarity or the reverse polarity and the designed set of landing gears can be fitted with magnets with opposite polarity. This repulsive or attracting property of magnets can align the propeller **204** in the desired direction within the enclosure formed by the landing gear **200, 300, 400, 500** and **600** locked until the rotors **154** are electrically rotated again.

[0065] In an embodiment, a front portion of the enclosure formed by the landing gears **200, 300, 400, 500** and **600** can include a shape that diverts the frontal airflow in a streamlined manner, away, either above or below or both, of the enclosure,

[0066] Thus, the present disclosure provides a simple, easy installable solution to in the form of the landing gears **200, 300, 400, 500**, and **600** to overcome drawbacks of the conventional aerial vehicles. Specifically, the landing gears **200, 300, 400, 500** and **600** can be moved, after the AV has vertically taken off, to a retracted position in which the landing gears **200, 300, 400, 500** and **600** form an enclosure around the propellers **154** to reduce aerodynamic drag, Reduced aerodynamic drag can make the aerial vehicles more efficient and increase their endurance.

[0067] While the foregoing describes various embodiments of the invention, other and further embodiments of the invention may be devised without departing from the basic scope thereof. The scope of the invention is determined by the claims that follow. The invention is not limited to the described embodiments, versions, or examples, which are included to enable a person having ordinary skill in the art to make and use the invention when combined with information and knowledge available to the person having ordinary skill in the art.

[0068] A general advantage of the present disclosure is to provide an improved aerial vehicle that overcomes drawbacks of the conventional aerial vehicles.

[0069] The present disclosure provides an aerial vehicle with reduced aerodynamic drag during flight.

[0070] The present disclosure provides improved landing gears for aerial vehicles that reduce aerodynamic drag during flight.

[0071] The present disclosure provides improved landing gears for aerial vehicles that do not obstruct field of view (FOV) to the camera during flight.

Claims

1. An aerial vehicle (**100**), the aerial vehicle (**100**) comprising: a rotor system (**150**) comprising a set of rotor arms (**152**) and a set of propellers (**154**) coupled to the rotor arms (**152**); the set of propellers (**154**) being configured to provide a lift force during a vertical take-off of the aerial vehicle (**100**); and a set of landing gears (**200, 300, 400, 500, 600**) movably coupled to the rotor arms (**152**) for movement between a deployed position, in which the set of landing gears (**200, 300, 400, 500, 600**) support the aerial vehicle (**100**) during landing and take-off, and a retracted position, in which the set of landing gears (**200, 300, 400, 500, 600**) form an enclosure (**210, 306, 408, 508, 608**) around the set of propellers (**154**) to reduce aerodynamic drag during a forward movement of the aerial vehicle (**100**) after the vertical take-off of the aerial vehicle (**100**).

2. The aerial vehicle as claimed in claim 1, wherein each of the landing gears (**200**) comprises a pair of L-shaped support member (**206**), one on two opposite sides of the corresponding propeller (**154**), each of the support members (**206**) comprising a vertical leg (**202**) and a horizontal leg (**204**); wherein the horizontal leg (**204**) of the two support members (**206**) is pivotally coupled (**208**) to the corresponding rotor arm (**152**) at a free end of the horizontal leg (**204**) for movement between the deployed position and the retracted position such that in the deployed position a free end of the vertical leg (**202**) of at least one of the two support members (**206**) supports the UAV on ground, and in the folded position the vertical legs (**202**) of the two support members (**206**) is disposed horizontally under the corresponding propeller (**154**) to create the enclosure (**210**) around

the propeller (154).

3. The aerial vehicle as claimed in claim 1, wherein each of the landing gears (300) comprises a C-shaped cross section comprising an upper horizontal portion (302-1), a lower horizontal portion (302-2) and a vertical portion (304); wherein the upper horizontal portion (302-1) is pivotally coupled (308) to the corresponding rotor arm (152) at a free end of the upper horizontal portion (302-1) for movement between the deployed position and the retracted position such that in the deployed position the lower horizontal portion (302-2) supports the UAV on ground, and in the folded position the vertical portion (304) is disposed horizontally under the corresponding propeller (154) to create the enclosure (306) around the propeller (154).

4. The aerial vehicle as claimed in claim 1, wherein each of the landing gears (400) comprises a C-shaped cross section comprising a pair of vertical portions (402) and horizontal portion (404) connecting the two vertical portions; wherein the vertical portions are in sliding engagement (406) with the corresponding rotor arm (152) for movement between the deployed position, in which the horizontal portion (404) moves down to support the UAV on ground, and the retracted position, in which the horizontal portion moves up to create the enclosure (408) around the corresponding propeller along with the two vertical portions.

5. The aerial vehicle as claimed in claim 1, wherein each of the landing gears (500) comprises a plurality of U-shaped slats (502) coupled to the corresponding rotor arm (152) at two free ends (504) of the U-shaped slats (502); wherein the plurality of U-shaped slats (502) are configured to be moved between the deployed position, in which the plurality of U-shaped slats (502) shrink to move away from the corresponding propeller (154) and a lower end (506) of the U-shaped slats (502) supporting the UAV (100) on ground, and the retracted position, in which the plurality of slats (502) extend to create the enclosure (508) around the corresponding propeller (154).

6. The aerial vehicle as claimed in claim 1, wherein each of the landing gears (600) comprises a plurality of flaps (602) connected together with at least one joining mechanism (604) and located under the corresponding rotor (152); wherein the at least one joining mechanism (604) is configured to move the plurality of flaps (602) between the deployed position, in which the plurality flaps (602) are oriented vertically with gaps (606) therebetween to allow airflow to pass there through and also support the UAV on ground, and the retracted position, in which the plurality of flaps (602) are oriented horizontally without any gap therebetween to create the enclosure (608) around the corresponding propeller (154).

7. The aerial vehicle as claimed in claim 1, wherein the aerial vehicle comprises a first actuation mechanism to move the set of landing gears (200, 300, 400, 500, 600) between the deployed position and the retracted position.

8. The aerial vehicle as claimed in claim 1, wherein the aerial vehicle comprises a second actuation mechanism lock the set of propellers (152), before the corresponding landing gear (200, 300, 400, 500, 600) is moved to the retracted, such that blades of the propeller (152) are aligned with a flight direction to minimize a frontal area of the enclosure (210, 306, 408, 508, 608), thereby minimizing aerodynamic drag.

9. The aerial vehicle as claimed in claim 8, wherein second actuation mechanism is selected from a group comprising a mechanical actuator mechanism, an electronic actuator mechanism based on sensors, and a magnetic actuator mechanism.
